

ActInf Livestream #033.1 ~ "Thinking like a State - Embodied intelligence in the deep history...."

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[Music] all right hello everyone welcome to active lab live stream number 33.1 it's november 23rd 2021. welcome to the actinf lab first-time participants and returning participants and listeners we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is a recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams we are in the final two weeks of november 21 discussing thinking like a state paper of choice and we have some tentative paper selections for number 34 and likely we'll um take a break and start with 35 in the beginning of january 22 but we're really looking forward to an exciting 2022 so if you have any ideas for people to invite for other streams or if you yourself would like to join then just let us know today in active stream number 33.1 we're going to be learning and discussing this very cool paper thinking like a state embodied intelligence in the deep history of our collective minds by aval aka serval gwen and khalif and it's just a discussion video even though we have the author on the stream which is awesome we're pretty much in the dot one going to go over anything that we want to we can return to some keywords and bigger questions that we raised in the dot zero we can add notes for next steps for questions to think about over the coming week to touch on again in the dot 2 or possible ways somebody could continue this avenue of research and we will be discussing this paper in next week 33.2 okay we'll uh go around and introduce our panel today everyone can give just a short check in they can say hello and something that they remembered from the paper or something that they were excited about to discuss and then pass it to someone who hasn't spoken and then we'll conclude with the author so i'll start i'm daniel i'm in california and just as we were discussing right before we started i want to explore this aspect of states as anticipatory cognizers not just reactive cognizers and i'll pass it to dean hi i'm dean i'm in british columbia and ensconced in a uh cedar box today out of my cabin and i like the paper uh

because i want to see whether or not there are parallels between what happens on an individual level what happens on more organized levels of individuals and i'll pass it over to blue good morning or good whatever time of day it is in your world everyone i'm blue knight i'm an independent research consultant in new mexico and i like this paper for many reasons but my favorite reason is the scale free aspect of self-assembly so it's self-assembly at a much larger scale than i'm used to i normally study cells and sub-cellular organizations so i like how active inference unites the things that i like to study with statehood and larger scale organizations and i will pass it over to first time participant andy hello everybody my name is andy i'm in uh richmond virginia i'm very interested in looking at the mechanics of how uh intelligence works through a collective right uh are there universal things that we can that we can apply for just the movement of information through um relatively close systems and and how to modify that the cognition that goes along with it now we'll pass it along to stephen hello and good morning steven here in toronto um yeah i'm interested in this paper in terms of the scale the social uh impact of of of looking at states and how active inference scales are and also maybe a little bit provocation in terms of like how much are we looking at the the the dynamic states of organisms or macro organisms how much we looking at the state of a dead set of artifacts that helps us work with our niche and the scale of working with macro artifacts in the niche as much as macro cognitive agents so i'll be interested about that um and i'll pass it i believe to daniel thank you steven um i think we've all we non-authors have gone so serval would be awesome just to kind of recap hello and what brought you to writing this paper as we kick off the discussion so hello my name is aval and arseval i'm training physics epistemology um great things and uh what i'm interested with this paper is to show that we if we accept a naturalist understanding of recognition which is a saying that is based on actual systems actual dynamics then we can't resist the conclusion that social systems are organized in a way that is cognitive and i found this uh basic constellation pretty important because then there are no limits like systems that are cognitive uh work in the way they are structured to work and we can't uh we can change things and we can understand how change will affect how things behave which is pretty good jones because i think it's pretty likely that the national state systems that is dominant today nation state capitalism system is uh dying and is uh basically bringing us with it so we have to uh explore the state of alternatives pretty quickly and efficiently and this uh needs a reflective understanding on uh what this database is great and i think it's um it's a a pun or a homonym that will return many times what are the states because we talk about partitionings of states a lot in active inference internal blanket external nested

systems and now we're talking about the nation states of how are social systems cognitive we use that state-based partitioning to talk about other cognitive systems ones that are designed or evolved and so we're applying active inference to study the cognitive properties of states um steven and then anyone else and i'll just keep on adding notes or questions and people can look at it or suggest things in live chat go ahead stephen thank you yeah i think it's also cool with active inference there's that kind of ontological challenge whereby we're dealing with non-equilibrium dynamical states in a nonlinear system which gives rise to pullback attractors which have this non-equilibrium steady state quality where however often states are or say we're thinking quantum theory or something you know they're like it goes from this particular state to that particular state right so um there's there's also that interpretation of what can be done with states when you start to have non-linear dynamics at play thanks blue so something that i mentioned uh in the dot zero video that i'm interested in maybe exploring a little bit deeper is like what do states aggregate into so what is like the grander scale if we're all just self-assembling into something and and the quantity of states that are produced i can't remember who said it maybe it was a veil that states um like arise as a result of war faction and division and so as we are progressing forward in time there's an increased amount of faction and division and like it doesn't matter how homogeneous a population is they will find something to divide about and so like this creates more city-states within city-states within city-states so is there some kind of equilibrium like attractor space for like the optimal number of states or is it all going to be like unified or is it all going to be dissolved at some point thanks steven actually that's a good point there blue there's also something interesting about democracy they say that for democracy to work it almost has to be a split of around to 50 between different opinions parties otherwise like if basically one party is supported by 65 70 of the people generally speaking it doesn't work very well because then there's no there's no dynamic i suppose to move backwards and forwards um so i think that's quite interesting so when we talk about the minimum number of like states or or entities to exist for democracy to function opinions um it's interesting you know in the system that mike levin works with back from our july live stream the computational boundary of itself he works in planaria which are regenerating organisms but you have to have a minimum number of cells and particularly like a minimum number of stem cells so i think it's like a thousand cells at base and like 120 planarian stem cells or you don't get regeneration and so like what kind of dynamic is required or how many people and even for like you talk about hunter gatherer organizations like 50 to 500 people they think made like an optimal hunter-gatherer colony like less than 50 and you're

going to be eaten by coyotes or whatever so where is that optimal number in statehood what does that look like yeah what aspects of the system are necessary sufficient incompatible at odds with certain kinds of regimes of state and it is really funny we talk about policy selection policy sounds like it'd be more from international policy than it would be from reaching for your cup of water and then we talk about regimes of attention usually people hear regime they're thinking some historical regime of a state whereas regime of attention doesn't necessarily carry that connotation so linguistically there's a lot of contact points and that's why i think it was a awesome contribution by cervel was to just draw those out ground it in the physics of critical systems and of cognitive systems and that's what is contributing to hopefully some of the ways we'll explore it's going to be used going forward so dean and then stephen yeah this this might sound like pushback or challenge or whatever but i it's just i have a personal issue with the entire idea of scale free like that there's some sort of invariance across scales and i i think that like when we see for example a lower case a morph into an uppercase a even though they are of the same categorical type and fit in the same place in the alphabet there is a there is a change that goes on and so there is a scale variance aspect to this that i think so again this is my my sense of it as opposed to sort of the agreed sense that there's a scale-free process in play the other thing i would i would wonder a little bit about is when we take a constructivist view a city as built i'm not sure what happens to the bi-directionality that active inference affords so again i i'm not saying that we can't find qualities in both but i'm not sure that a city as a reflection of something constructivist is necessarily how living things reflect and predict but again i'm open to having my mind changed thanks dean stephen and then all have a comment on the constructivism i was just also i think it's a good point we blew about the uh mike levin's work and and the question of um yeah how things are scaling up and of course the big thing that mike levin did mention that you know they they've got these quite a lot of ideas around how things grow multiply and relate but no idea how they stop and that's kind of also interesting in terms of you know when we think about scale as like has the organism reached its point because the top scale the maximum scale has kind of been reached or is there some sort of is it that there's it's the dynamics at a mid scale which is like an action so it's not it's not that it's ever arrived at this thing it's just arrived at this meso action policy landscape which is um available uh and you know is that availability then understood from what's inside or is it something that's checked to something external in the environment so that that i think is an interesting question thank you um capitalizations yeah you have to the state capital capital punishment there's so many um other funny word plays but to the constructivist

comment one sort of simple constructivism would be like i mean there weren't bricks there and now there are so the city was constructed and now it's the scaffold for us to interact in so you ask dean where's that bi-directionality so i think there's a few ways to talk about the bi-directionality so first is niche construction can be a symmetric relationship we talk about the agent modifying the niche and the niche is modifying the agent and so it's like you have the blanket as the interface and then there are action and inference on both sides of the blanket being used as an as an interface and so it's more like a co-constructionism but if you know lego set a is building b while b is building a that starts looking more like interactionism because it's not just instructions put the brick here it's a specification of a protocol and an interface for interaction and that's what opens the door to improvisation into other kinds of dynamics that aren't just simply like assembly of the 3d printer and then the other point and then dean is it reminds me of dennis book darwin's dangerous idea where he talks about basically two ways that you can talk about the appearance of design in natural systems which are cranes and sky hooks and so if you look at a skyscraper you think how could that have appeared one approach is they built a crane to put that last top piece on and if that crane was too big to construct then a smaller crane built that crane so you could have a cranes all the way down like small things were built and scaffolds were in place that were removed later to result in this skyscraper or you can just cut the knot and just say well maybe there was a sky hook and they just latched it to the sky and then built down so that's like top-down constructivism appeals to skyhook technology whereas bottom-up emergent process-based co-constructivism is more of a crane based approach so i think that's a very good point how even with constructivism we can still retain a lot of the important aspects of constructivism like physical construction but we don't need to fall into a uni-directional um or a top-down trap so dean what do you think about that and then stephen or anyone else i think i think the important point daniel is what you just mentioned which is the trap of prematurely collapsing to the top-down privilege um for example uh if you go to if you go to just outside of denver you end up in a place called red rocks and then the question becomes did did uh did somebody enact red rocks because they put a stage at the base of an amphitheater or did red rocks enact everybody who was attracted to that place and so all i'm basically suggesting is is is when we look at how things are constructed we don't want to lose sight of the fact that sometimes things construct us as well sort of the sky hooks aspect of it thank you dean stephen yeah and also i mean cranes and sky hooks it gives that idea of how it links in they're also dead items right so they're both tools or they're both objects um so and we can we can bring in sky hooks which might be

informational instructions or cranes are you know places to carry the bricks that are going to build things but i think that when we talk about that markov blanket and what goes both ways is yes i think when we're dealing particularly with dead things like bricks and that we construct an environment now it's a bit more complicated when we impact an environment that also grows and changes and is alive however when something comes back the other way from the environment is it constructing us or is it scaring us or is it giving us lots of little punches in the gut in which we heal so i'm wondering whether the other direction it can't construct us but what don't kill us makes us stronger type of thing um andy please and then blip dean i think in the context of cognition in the state the specific um the element that's missing right is what's the relationship between the nation and the state right so the nation would be the bottom-up information or the bottom-up cognition and the state could be theorized now as the top down and then broadly speaking right the nation creates the state the state the existence of the state changes the circumstance the changes in circumstance forces the nation the nation the people to rethink how they're operating in the world around them because their circumstances have been modified and now you have this constant bi-directional relationship between bottom up and top down in the context of state versus nation and modifying circumstance over time blue dean go ahead oh yeah dean yeah andy uh i like what you're saying except it's already collapsed to an a to b relationship and i think what active inference allows for is to keep things up on it on a generalized abstracted level it doesn't have to be here's the nation a and here's the people b that's the difference essentially is if you want it to be an a and b thing go ahead that's that's good for argument's sake but in fact what active inference does is it keeps one side of this open and not necessarily specific it keeps it in order to be predictable it asks the counterfactual what if as opposed to having that pre-presented as the relationship and it's new it's a new type of relationship that's arising and i think that's right explored in the paper in the deep history section there's really the specifics there's the minute particulars of the specific states that we have and typified by westphalia and a bunch of other historical events that the specifics matter we're in a complex system the trajectory really does matter so i totally forgot what i was gonna say but i would like to maybe challenge andy like what is the difference between the nation and the state like is the nation like the people that make up the state and then the state is like the collective of the people like is it like a a part whole concept or are they two separate uh larger entities and people make up both words depending on which definitions you go with right it gets a little bit squishy in the context that i use them state is a construct the political organization of a people is downstream of uh the

shared identity of the people of a people so the nation would be broadly the largest sense of shared self distributed among a relatively close system in this regard and the state being the construct by which these people organize their affairs inside of their nation that's a relatively new idea right the idea of a nation and the nation state is relatively new as far as like an overt construction um beforehand in in the age of absolute monarchy there was no separation between the state and the people right it was all embodied in the in the person of the sovereign different different historical contexts different periods of time different cultural context now you might get uh different definitions but in the way in which i was using it the state would be a policy requirement of the culture in which it was born right it's why american democracy functions significantly better than south sudanese democracy even though they have literally the exact same institutions so daniel maybe you want to flip to slide um 33 i think like where we we like kind of show in the paper all of the like things that go into like the production of a state and so i like i'm curious as to like where nation fits into this um is it like the in the collective identity or is it like a relationship between like nation and collective identity or just what your thoughts are on that yeah absolutely collective identity to me would be the definition of a nation shared sense of self everything else are things that you may or may not need a state to accomplish that may or may not be more efficiently conducted in the auspices of the state right um control the monetary flow you probably need something as sophisticated as a state but to transfer resources back and forth you don't right you can you can grow grain and you can you can trade it for for me uh go ahead well just the the um collective identity i'm just thinking about in organism and the liver and you know the bone cell it's like collective identity doesn't mean identical phenotype and so it makes me think of a cultural memory where each person maybe there's a few touchstones that 80 percent of people know about or increasingly though there's less and less shared information amidst the informational changes that are happening so what does that mean when everyone's only going to have a little partial facet of the collective memory on board but maybe they can enact broader processes that are truly collective so how do we respect differences shared aspects across multiple levels so um blue and dean and then serval want to hear what you're thinking about this so just like daniel how you were talking about like the bone and the liver cells like making up an organism um you know i think that that speaks here to what's um the specialized roles and like the density of settlements enables that like functional specialization and like you can have a blacksmith and a shoemaker and a farmer and everybody takes on their own little role and then goods can be redistributed so it just like reminded me of that and like that functional specialization um

and the importance of it thank you blue dean and then cerval or pass yeah when i was watching the 33-0 and i saw this slide thank you blue for asking for it to be um brought brought back again here this is again an uh a specific to a specific because it's historical and it's reflective the fact that we can look at things and say this is what happened we can look back on them i'm not sure we want to necessarily conflate that with active inference which is about the recursion this is what happened we can see how things moved across a timeline from point a to point b a city state evolved but is that recursion which is what active entrance is about which is variation upon variation i'm not again i ask is that i know how we can i know how we can diagram it and represent it but is that representative of what actually occurs in recursion when certain thresholds appear and then just as quickly disappear interesting so serval if you'd like to add anything and then stephen so there are a lot of threads that were opened what are you asking my opinion about i think part would be just uh on nation and state similarities differences um what is and what if and then more on this sort of integration slide where does active inference come into this tying of the knot how does acting in france uh about nation i don't think it does nation is simply a collective identity that is hated to a political entity it's community of peers that happen to have some stakes and power over the behavior of the institution material states and the fact is that nations are not universal you have a lot of states that are not nation states that are not suited to a community of peers for example the roman empire it was a empire which means that you have people who get the power that form authentic group that is the roman and you have all the rest of the people that exist in there that also are ruled by the um by the empire that is dominated by the romans so you do not have a community of interests that illustrated to a state you have a netting group that dominates other ethnic groups through a state the nation are not um they are not invariant aspects of what the state is however they are um a collective identity or at least ways to build boundaries between a group and ad group there seem to be an environment of efficient human collective action according to astrum's principles there may be more tendency to build nations to the nation states uh when you have more uh importance uh between group selection it is said that i don't know of any uh computational quantity work that uh but he said that modern europe is pretty extreme in terms of um between group conflicts of war the diamond agreement is that the geography basically enables trade at great scales but prevents the constitution of an empire in europe historically there were a lot of states in europe as compared as say uh in china where you have a big state that i don't know in terms of surfaces maybe uh two or three times like western europe is and more integrated than when you can

have in south america because of the mountains um so you have a pretty ideal regime to get a lot of intergroup conflicts and to get a selection regime that favors nations uh organizations that somehow collate with states and that's what happened to be instituted through uh i don't know westphalia and french revolution i don't know the story of that there are a lot of parkway pathways in europe from the states but in europe you got nation states originated you've got pretty aggressive kinds of imperialism that originated and both can be traced i think uh to an intense regime of uh between group selection and of war there are results that tend to say that war explains in the sense predicts more political complexity which is uh a debatable construct that basically maps this kind of integration of society the uh history of europe of modern europe specifically um and meteorology everything past the roman empire would have uh selected for both nations aggressive kinds of imperialism and this is the kind of organization we get to live in now serval just like the ecological regularities through evolutionary time scaffold the evolution of an organism's cognition this is a transposition of that argument to the biogeographical and contextual and ecological factors that give rise to and scaffold the development of states as things as cognitive entities there are some parallels there so it's an awesome point thank you blue and then stephen there is a pretty important things trying to say it that is a state does not actually correspond to a nation maybe states have original nations maybe they are assigned nations now but a state need another entire nation you know you cannot states like cultural population united states well that's kind of the usual deal actually yeah election is a collective identity in a community of interest states uh control systems uh built from administration and it can be and often is captured by an elites that is um technically social like socially in terms of cultural norms lobes pretty distinct from the people controls chicken in the egg egg as the secreted chicken as the life form but then there's a co-evolution so that paradox stands there too blue steven and andy i'm gonna defer and come back around to my idea so stephen and then andy yeah i think the the whole gale question is very interesting here um and um where it sort of so often i mean in the european union you've got the european union it's kind of a reunion of pluralistic partners is the idea and you've got this tension where one side is basically saying no you're not you're a state and we want to be a nation an independent nation there's this tension right between oh no we're a union we're a union and when actions are working that works the union right and uh so i think that there's a scale where people try and work at and and the rate at which people work at what's the dynamics of people in the same way that when you have um politics seems to have a very slow trajectory you know things shift you have the state you have your end identity of your

party however nearing an election a week this praise is a week is a long time in politics and it is near an election the dynamics start to become much faster things can change much quicker so this scale question again comes in is what is what is a unit of identity how does it manifest at different rates and just one other thing in terms of identity is in participatory community-based action research okay they talk about community is actually a unit of identity it's almost like there's so many different types of fluidity at the community level and overlaps between communities that it becomes a unit of identity to actually work with in measuring stuff we tend not to do that like at the higher level the nation it's just the flag is a unit of identity it's a very which hasn't maybe changed in 100 years so it's an interesting um as well as the you know what how much can things change and how much are the ability things to change caught up in the artifacts that embody them the artifacts that constitute them in some case so and any comments or to blue yes thank you um i was going to well let's let's go deal with what you just said right community smallest action uh element of action in this context uh identity is the smallest element of action however it is guided or bounded right by its ability to affect change so if you if you deeply identify with one other person in this world the two of you uh are the small that you have you identity becomes basically shorthand for interest we two people by calling ourselves a group have a certain worldview and a certain desire to see our interests achieved we have the capacity of two people to accomplish that so what is our ability to affect change at that scale relatively small once you're talking about anything larger than interpersonal relationships right but it doesn't change the fact that your identity is shorthand for your interests and that we all at whatever level or scale we're operating at are pursuing our interest to the best of our ability bounded by relative power right um but this idea of of nation versus statehood survival you're absolutely right the state is a thing independent of a nation sometimes they correspond and typically we say that that is um that that's a good fit because otherwise it's an empire right otherwise it is somebody else's ideas of what constitutes uh right or wrong in terms of projected values or institutionalized values that are now imposed on you the state is not representative of your cultural world view or your shared identity it is something provided to you by somebody else's in the form of empire but the question that i had right going through the slides there was this idea that the state as a top-down hierarchical control system what is then nation or what is an identity how does that function the sharing of information and cognition relative to the state thanks for the uh text-driven question it was a core argument of the article just to quote that states can be and should be understood as hierarchical um avail if you want to give a direct response otherwise dean in blue so there is a

direct response uh which is that uh to talk of hierarchical control systems specifically to talk after done control this is i think i've reframed the same uh point the same thing in saying those both uh words and of course there is no cognitive systems that is purity of thought but there are they do not work they live in labs and they are not useful uh so basically uh the organic organization of whoever happens to have a say in what the states is is the at least the ascending dynamics in the collective when they are what affects uh in bears in terms what uh change the priors and the policies etcetera in the case where you a defined nation a defined entry group that happens to control the states which i'd say is exceptional from what i've read you can't say that there is a meaningful nation states fit and the nations basically the brain and body of the state in the sense that well the ascending dynamics that influence what the control systems does in most cases you will have i'd say a pretty much stronger decoupling because basically only elites would get to say what the state does and in cities where in mercedes the elite is not to define an ethnic group in that case you will not have a clear fit between nation and states although of course the committee of power the defender states if you conceive it as a nation will uh define what the state's prior well pacified to define because of course there are the tools that register thank you so awesome lot of hands so dean then blue steven um so i want to just circle back to what andy just said andy when you if if you go and look at this live stream three hours from now even when you see a perfect reflection we'll assume that this youtube video comes out exactly as as it intended i think what you just said proves that we're not talking about scale free talking about scale friendly because what you will see is not what you just did and so i think that that has a huge bearing on this idea of what we're talking about here we can shape-shift into what cilit has pointed out here is that that fact that we can fill between these two levels of scale doesn't mean that it's it's it's one size fits all it quite literally means is that we can fill we can change we can modify and i think that's recursion in its essence as opposed to taking a reflection and knowing that the next time we look at that it's going to be different because there's some elapsed time between the moment we set it and then the moment we heard ourselves say it dean i think it's an awesome point with scale free and scale friendly skill-free people usually mean to say that the mathematics or the formalism doesn't have an a priori system of interest or scale like this is a centimeter theory or this is a kilometer theory so it's scale free because it doesn't have that a priori reference scale but then scale friendly it's almost like how about you tell me what scales you've identified and let's make them friends through interaction and iterated modeling and so we're friendly to scale and scaling but that doesn't necessarily mean it's free although from the other side it's also fair to

say that it's a scale-free framework in the way that a bayesian computational framework or even a linear model is scale-free it doesn't a priori say that kilometer is the default unit for linear regression either blue and then stephen so i think what i'm going to say is going to touch on the comments that were made by both andy and cerval um first of all in terms of this relationship um like the mutual interest being the fundamental unit of identity and like collective identity this really goes back to expansion of your cognitive boundary beyond yourself and like when we were talking about the computational boundary of itself again i'm gonna like always refer to that live stream but um there's an informational sharing when there's an identity sharing so when you expand your cognitive boundary to include more than just yourself like if you have like a really deep connection a really deep relationship with one person you have like a a great degree of mutual information between you and that one person and then so say you have like a family unit or like a network of close friends there's a lot a large degree of informational sharing and so in this sense of building a nation um like the fundamental unit of identity then maybe is is actually informational so it goes back to like bayesian principles where there's this informational sharing that that then allows you to update your generative model um so so just to kind of speak to that and then also maybe um just like maybe help andy make the link like between um downward causation imposed by like a government or like a hierarchical control institution and like the the emergence that happens like when people come together and share information in a collective there's a super cool paper by david krakauer called the information theory of individuality and it talks about like in order to become an individual and there are many different kinds of individuals but in order to become an individual there has to be like a bi-directional information flow that's both emergent starting with like the fundamental units like the people sharing information there's an emergent and then a top down so bottom up and top down and like when you have this bi-directional information flow that's like what allows you to form a boundary i guess around your collective something like that thank you blue steven and then anyone else yeah i um so picking up what dean said there and um and actually blue and everyone else is the formulism is is scale free at the free energy principle and then you get into applied active so then you get into which has a sort of scale-free quality and then you get into adaptive active inference and then you get into you know inaction of adaptive active inference and then you get organisms like us that want to apply active inference and i think that i think it does like really start to go scale friendly once you start to go up um and and one thing that i find interesting i've been this actually is nice it ties into this infusion space work that i do but i look at the face and the body

so at there's a particular scale at which my mind is embodied and you can see it and the bit that you see is the bit that we show on this screen you're not looking at my foot or my shoulder or my hands you're looking at the face so the face is the embodiment of my mind as an identity okay this is my projective identity and it becomes very useful so it's it's scale friendly at that scale but you know until the other scales of my body aren't you know my liver doesn't do a great job of my identity okay and then we got okay what scale do i have to get to i'm talking around if i'm just talking one-on-one i don't sit here waving a canadian flag or whatever right now if i'm a crowd of a thousand now maybe we need to have a big flag on a stick right and that becomes my face right that becomes my projection but that's a scale friendly because now i've again it's kind of gone in and recapitulated the environment so that's kind of interesting um thanks thank you stephen cervald i don't really get the debates about uh scale fiscal friendly thing uh because what a scale-free theory is is a theory that is in principle a formalism that's in principle it's not a really different scale so for example my opposition to um the weak and strong forces that only have a measurable effect at very small special scales uh gravitation and electromagnetism they can influence things but that are pretty much so far well not release money some because you have the um cons mutual counseling of plus and minus charges but in principle it could affect things at distance because of uh the error square the repentancy just kidding and uh the fact that you have a theory that is capri does not entail that every system where you express the theory is calculus for example you have the gravitational theory the skeletri but if you study the solar systems it has a scale which is about the distance from the sun and the uh thing apart if you study electromagnetic motor it has a scale which is uh well the size of the matter and if you study a biological organism with a scale pre-formalism it has a scale which is whatever scale at which the organism happens to exist so i don't get what the debate is about how about give a thought on that yeah i got a thought on that so i think we can see what a reflection is we can see what a triangle is we can see a very small triangle and we can see a very large one those are reflections but that is not recursion and as as soon as you get into variability upon variability you're going to have to come up with something more than a reflection and i think that's why you have to come up with something that more than i mean we can see scale free and nobody's arguing against that but what was i guess what i'm pointing out is that if we limit ourselves to that we're not going to see the entire picture all kind of intermediate i think they're both very valuable views scale free is certainly more aligned with the way that people do talk about formal models whereas scale friendly is a little more of a dean-ism which is always fun i see skill friendly as emphasizing the active role of the modeler

in connecting the levels so it's not enough to just say like i have a stretchy ruler that could apply to different scales it's not an a priori scaled ruler and then making it more of this process of approaching the mechanisms and the processes that are interacting amongst scales rather than just thinking well we'll put the linear regression on a little thing we'll put the linear regression on the big thing and then we'll have a scale-free applied theory this is more like let's take that system as a indivisible unit of function and then like interact with it in a little bit more of an active way and it can still be scale free for those who like that way of talking about it um but for those who want to get to know it maybe scale friendly is approachable as well there's also active inference and adaptive active inference and inactive influence so i think like daniel made a good point there is keeping the active influence flowed through and helping that is really useful right so that we don't lose the formalism side because that's the danger right we clip off the but but there there is something for instance at what point is adaptive active inference possible like is that the point when you need to have the brain within a reason or there needs to be there needs to be some generative process that's regulating active inference for it to then adapt at different temporal scales um now that that raises some other questions about exactly where but there's that you know even the idea of what adaptive active inference and active active inference or adaptive and active potentially is starts to then tie in well you know and if i can only do certain adaptive active inference with the brain then has to be the scale of an organism that has a brain and below that scale it can't so there's something that becomes transitionary or contextual dean on this point and then to serve all for a combined response yeah um so again i think knowing that a triangle is a triangle regardless of the volume of its footprint is is important i think when we include updating when we include time then we should include scale friendly so i'm not discounting but i'm saying that the updating process whether you want to call that adaptation or in action the change aspect of it doesn't get resolved simply because a triangle is a triangle if that makes sense bucky fuller said angle is is metaphysical and general it doesn't refer to any specific substrate to say that something is a 60 degree angle and frequency is always bound up in space and time and the specifics so any realized system is going to be both serval so um stephen i think talk about an active acting in france rtx in france is uh something that keshav introduced in 2018 i think something like that uh tries to reproduce inactive notions of cognition from active influence what it says is that you have agency when you have a hierarchical modulation of active influence system you have higher level of action fronts that modulate lower levels of active influence the question it rise is uh basically a question that is not solved and i don't even know what's happening on it

in the ontology of science and naturalism which is do you have different level of existence or do we have only scales and even at this level when we admit that we have things within things within things within things and that's what the world is it's complicated the thing is that uh to circle back to a question that has been raised initially about how what in france relates to other values of the while keyshot tries to formalize inactive concepts of cognition is entirely wrong because in activists they do not say organisms are heretically modulated acting from theorists say that the brain is killed modulated and they apparently projected its own activists but we say a system that is cognitive must be operationally closed which is it has set of it has a structure uh or an organization i don't know which because of their consumption uh that basically reproduce its own existence without any clear support without autonomously from its environments uh that's not how they say but whatever and you have a specific formulation that's computational it was formulated by um people from the basque center of life sciences i think the name is uh which is uh constrained closure so we have constraints that are slow processes that shape how uh faster processes behave and wealth a lot of them and collectively this lot of constraints shape faster processes so as to maintain themselves that is um constraint closure and that is what's the latest version of um on the elective um not agency but autonomy nation is and this entails not hierarchical modulation but uh meshed nets of processes that make something else exist at in between scales and the notion of what is the antics that choose to give this is pretty complicated i don't know i a lot of people that think that level is just an epistemic construct that is built by scientists and this is pretty consistent with a radical approach with a mentioned approach this everything becomes ambiguous the meaning of every single terms become vigorous so i have um no idea what will be a correct way to refer to this uh scale free says what it says that it is not interested in specific scales of behavior and uh it also says implicitly when we are in the context back in france that it is not interested in a specific antique level what applies for ourselves applies for a mutual organism which applies for societies and which in the case of vanilla in france also applies for dust vertex flames and whatever i'm not sure we're still and i'm still not sure what the national scale friendly is supposed to do and uh how it relates to uh the question of whether states are properly cognitive because we don't need a proper version of population of scales and proper anti-distinction axing distinction between what is uh recursive and what is only a scaling variance we only need to state what being cognitive is and to check whether states are and of course uh what this means for the broader uh episode constructs with builds around this thank you sir blue then steven so i have to like go back to um the active inference live stream number 30 we

talked about how to count biological minds and really this like raises so when you're talking about the distinction between adaptive active inference and like mere active um and in this this theory about how to count biological minds is everything that's biological is do all things then have a mind like even down to like the you know single cell organisms and in this in this paper like there are a lot of people who like have this like mind life continuity hypothesis and a lot of people really um ascribe to that so in this adaptive act of inference versus mere active inference i couldn't find the line like i really pressed on matt sims about this um and and he responded beautifully like he also couldn't find the line so at what scale or what level does adaptive active inference transition to mirror active and what i'm fundamentally getting at is like okay so are you doing mirror active inference if you're a single cell and you just move forward and back like towards a repulsion or an attractor like is this mirror active inference or is this adaptive also and so like adaptation to the niche like to me i think that happens at all levels but there was this distinction made in that paper that like adaptive active inference happens when the collective forms some like emergent behavior that makes it more fitted to the niche environment and so like where is the state then a biological mind does the state make um a set of organisms like more fitted to its niche through this like collective adaptive behavior or or is it maybe maladaptive because overall like as a species the more divides we have like the more war we'll have i don't know thank you stephen yeah thanks for this meshed nets idea i think that's that's definitely true um and uh let's kind of fix in with the kind of meshed architecture um i i wonder if like um blue was mentioned about this different scales of active imprints um identity is quite important here um the nation state um is a kind of is is maybe identity driven and i kind of wonder if we just scale that down i was thinking scale down i was talking about the face there earlier but for the baby in the womb okay is the the release during childbirth of the hormones which triggers the woman to break her waters basically like a broadcast from the identity of the womb so okay body i'm here now i'm a child bearing womb i'm declaring myself in the same way that maybe the pituitary gland but it does it all the time declares itself certain times to crash out into you know create insulin so could we see that then this meshed architecture comes at a scale where it's possible once the you know and then it's like and in the same way then uh the nation state maybe as we go up in the expression of uh signifiers to broadcast out becomes greater and greater you know becomes a bigger and bigger part of it and maybe the biological processes become less a part of it well definitely it's the question of realism and instrumentalism whether it quote is a broadcast or whether we'll just choose to model it as so one of those is definitely safe territory and the other

one is a little bit harder to make claims about what the territory is versus the map that we choose to draw around the territory um but very interesting points dean okay i'm one i just want to make sure i'll get feedback from people on this so to me adaptation okay adaptation around what is which can be scale-free or if we're talking about active inference do we also include the counterfactual what if and i think if we include the what if then i'm not sure that adaptation looks the same it's not reflective so i'm not pushing back on serval as much as i'm asking are we over fitting are we saying that recursion doesn't matter and are we saying that scale friendly doesn't matter because we only want to concentrate on that's my question dean i'll give it a thought then stephen and then anyone else active inference agents that are capable of doing adaptive active inference as a title so just a recently proposed word as serval brought up their internal generative model includes a model of perception and action as well as of cognition so that's metacognition the what is of metacognition at one level can be a what if at another level what is is the agent can actually be imagining a counterfactual with respect to perception and action so friendly scale it helps us bridge the what if and the what is because um there's also there's so many other angles one other path to walk through that would be like um in anil seth's book being you there's a great discussion about how counterfactuals actually are like the basis of cognition like how do we know that we're looking at a real mouse like on my computer table versus like a picture of one at the same resolution because actually there is an implicit counterfactual i could move at a different angle and see something different the physical object but i wouldn't expect the shadows to move on this jpeg and so actually the difference in our perception between two things that visually can look identical relates to deep situational awareness that actually blur or at least reimagine the relationship between what is and what if stephen and then blue thanks actually this paper's serving as quite a useful bridging uh provocation um because as you just mentioned there daniel with them okay so what if you know way of what if and counterfactuals so there's there's the cognition which is a particular scale of perspectival or even implicit what if the counterfactual but what if this and that happened in the future well the other question is what am i becoming and actually if we want to look at scale if active inference is based on free energy principles and free energy principle was pretty much the only form of nonlinear dynamical system out there but now we've got some of the statistics stochastic chaos work um also um like if you go right back to non-linear dynamical systems and a lot of discussion around non-linear dynamical systems people are talking about thinking about that as a process of becoming how because it gives you a way what dimensionality can use well the the process of becoming starts to tie into that

um and cognition as we see it doesn't maybe the process of being is more approached of becoming but the process of what i broadcast if using that term very brutally but what my identity conveys is a much more steady much more of a an artifactual state if that makes sense especially if i take a photo of it and put it on my profile on facebook thank you um here's an article on being and becoming and that distinction as well as i'm sure a panoply of german words has been discussed for a long time about the relationship between being and becoming and just to give one thought before blue that generative model of the active agent includes what kind of agent am i what are my expectations preferences affordances etc so we get in the deep active inference framework built in what if and then what and then who would i be and how would others see me and what would they think if i were to do this and then this were to change that's all part of the state space of what's possible these are like the grammar in terms of the very austere outline of perception cognition action and external dynamics that allow composition and new design motifs for those kinds of questions to be asked so if you wouldn't mind flipping back to the twisty slide i think it's 33. um and when you're thinking about like are we talking about what is or are we also including the what ifs in the model like when we talk about what it's like to build a state so so this is like how states have been built before but but how can we like take this and like as as daniel was like referring to at the beginning of the talk like how can we take the what has been and incorporate like what if via active inference to predict like how to optimally optimally form states in the future so is active inference incorporating the counter factials are we going into what-ifs or or like what about what's next like not just what if but but what's next to predict like the the cognition essentially of a state and how it's it's going to think going forward great question and what's next we can have our expectations is we can have our preferences they can interact in a different way so dean then andy yeah blue i think that ties in with the point twos that we tend to do all i would all i would all i would add to what's been said is that if if we're looking at as as is we're looking at this reflection and and can we ask off of that of what are we becoming of becoming which is a function sometimes of scale many times of variation so i know i keep beating this horse because i want to include the variation and the fact that time changes our picture that's that's all i want included in and i don't think that that as is a perfect copy as we update so i maybe i'm maybe i'm explaining this wrong but i think as change occurs that perfect reflection that perfect copy doesn't move forward without variation concentration sometimes sometimes we just toss it out the window andy that's the what if sorry thank you dean andy and then i'll give a thought on that as change occurs so uh in my travels i've run across this idea that somebody

tried to quantify functions of a state that are universal right what are the things like how do how do you maintain health care not necessarily which way is better but distribution of resources education health healthcare things that we associate with the state maybe not every one of them every state exists or every nation exists or let's stick with state maybe not every state does every one of them certainly not all of them do it to the same degree of competency and they are all bound within the expectations of the people who um comprise that that polity or that that entity but to get after your question blue right if we if we want to get to what if how is this different how does states form differently what is the optimal thing then i think you've got to get down to like um a core set of variables for which states are supposed to uh provide solutions to or address and then from there we can see well what if they did it this way what if if they prioritize this and not that what if they prioritize this function over that function and why would they right what are they responding to i don't think that they're responding to necessarily uniquely physical or material requirements i think there's some interplay between intellectual or values-based requirements as well as material requirements all mixed up in time and space given which parts of history we choose to preference in a particular moment so one note on the as change occurs here's a very influential book with a a broad follow-up lineage on the geometry of biological time by winfrey and it brings us back to the discussion of kairos and how change occurring in chronos we become like pro trustees we have to stretch and pull and change the sequence of action to fit onto the scale as defined whereas thinking about the space of action the geometry of biological time is defined more by action like embryological events rather than how many seconds have ticked on some atomic clock in some bunker that um as change occurs can be through time but it may not be um the kind of time that gets ticked out on a clock and that's just a very different way to talk about actions and inference because it's dynamic but it's not just like okay the clock's running and things are changing it's like there's actually a timeliness to action and so we're talking about the dynamics of the context and the timeliness and then maybe there's ways to map kronos and kairos or maybe one time we play more into one or the other so dean and then anyone else yeah just real quick thank you andy because i think what you did is you applied a partition to preferences and priority and you made sure that we didn't say that they were one and the same and so in real time we we see we see an evolution throughout the discourse so i think that's that's a real example of what kind of happens as we update so one other random thought just from earlier and then of course anyone just bring up a thought or anyone in the live chat so we were talking earlier about like capital with the ol and capital with a l okay so this was pretty

interesting capitol with an ol comes from capitolium the name of the temple of jupiter at rome on the capitoline hill dating in english from the 14th century so that is referring to like this building that's like very much like city on a hill and um city on a hill is used as a multi-scale allusion to the united states at the very least um like capitol hill but of course broader notions of what the state stands for and then there's capital with a l capital on the other hand has a far wider range of meanings in application the word comes from latin as well from kaput meaning head and then it has a lot of um polycymous uses like capital punishment which we talked of being the seat of government like sacramento is the capital california but then like venture capital capitalization so that's just interesting um i hope it's not simple word play when there's a homonym but it's very interesting that they both draw back to rome which was alluded to earlier when thinking about the seeds of what today's states are on the world stage so andy then stephen yeah sir would institutions of the state right like um ministry of foreign affairs can be considered an example of embodied intelligence in the context of cognition for yes something that blue said earlier people proposed a collective agency to be based on the ability of a certain kind of regime of her behavior influence individual fitness what i would say is that this is wrong the biological organization it is by definition organization it is constraint that exists upon the collective behavioral system if you look at say its steel beam it is a structure it is something that affects how the steel beam interacts with the world and this is something that is entirely invisible from the perspective of a single particular steel which is the natural lower level lower scale perspective you can articulate so the view that organization is entailed by things that affect the niche of their constituents is flawed what organization is is a constraint at a specific scale and specific behavior on a specific dynamic the relation it does with the body the body certain uh certain schedules of cognition is about is uh the sets of those constraints uh that collectively enacts uh behavior that we say is cognitive so when a ministry of foreign affairs or something is doing a thing it is embodied in the sense that it participates uh in a greater network of constraints uh which collectively build organization which entail the behavior the specific h-specific decision of the um ministry of foreign policy so for example if there is some kind of panic moral panic against communism it's likely in the absence of any uh formal statement against communism uh decision be taken by that this shouldn't be best against uh specific people and words and themes but urban needs communists and this is not entailed by anything at the formal level at the level of symbolic computation that could be construed as competing the highest or most recognizable forms of state cognition yet this participates in a collective behavior that's adapted from the border system this is basically the

nation states distinction familiar the aggregate behavior of specific sets constitution constraints past processes will build the strong processes and physical entire behavior and this is not something that needs to be acknowledged from any higher level and this yes is uh cognition and this grounds of view of hemorrhagic recognition are something that follows states um as something that is uh grounded in not the body of any single individual but in the body of the state as the set of its consulting constraints thanks cervel i also put up a quote that people can pause the video and read from the gulag archipelago that specifically refers to organizations as organs of the state which is a little bit of a almost sinister meat grinder way to describe the state and it refers to this interplay between the awareness and the actions of individuals and the systems outcomes at the state level yeah i think as well as this this is really useful i think this embodiment and our intuitive ability to embody um does then recapitulate the influence process so as well as these terms the seats of government or whatever coming from say we see it coming from roman literature but over time it's been encoded and encoded and maybe abstracted out and we use words and sometimes you have to look up the etymology to sort of find out where they came from but we we do intuitively see a difference if someone says someone's going to take their seat at the senate where you get a feel for that but you don't say they're going to take their seat at the palace they're going to take their they're going to they're going to sit in the throne right because the whole point of the palace is that those seats are not everyday seats right so there's there's a but you get that because of how you know space is treated around one of those seats like you don't you don't put a load of seats around and roll up a load of thrones you give space to it um and there's actually a lot of interesting work in physical theater where you literally can work with space to convey how much space you give someone will convey how much power they have that reminds me of the term meronymy so that's like when you say there were 10 mouths to feed it's referring to a system by referring to the sub part of the system so it's kind of like marrow topology is about the parts and holes and neurology and so the meronymy is when revealed through our language we say there were 20 seats on that committee and the fluidity at which that is semantically unpacked shows that there is a relationship like when you say there was 20 mouths you think there was 20 people and that includes a lot of our assumptions um you know 10 pairs of hands or you know 10 hands what does that say about our assumptions about you know ableism or about what the default in our um way of speaking like that stephen and then anyone else yeah uh and well ableism definitely comes in then as these these codes as people start to look at ableism that that's a very good point and also there's that intuitive understanding that you know there's the

idea that a team generally is really more between five and ten people to optimally be a team and then you've got this idea of a group so you have an intuit when you hit 20 you've gone past a small group into a large group and there's a whole work around large group interventions and getting aspects of the system in the room which is about when you can no longer look at each person in the eye and hold it which is sort of after 15 16 people so again you hit this to some extent we don't need to be taught that these things evoke the reality for us as this structured organism in space in place now it's probably different for a dog on four legs maybe they can only sniff around four dogs right and that's a big crowd for them so you know and is that about their mind or is that about their their morphology so is this definitely an interesting thank you serval and then anyone else i like the way you talk of the throne as something that uh whose function is communicated through medically embodied use the community activity of the state is embarrassed in constraints that are not human but uh those constraints needs to somehow affect the behavior of humans of because those fact faster dynamics are basically the dynamics of a human individual they need ways to constrain it and influence it to exist as constraints a major way they do so is by building the niche humans living so building the way humans embody their perceptions relate to each other to each other take decisions etc etc to simply build a monumental architecture to convey that specific buildings have specific functions and are very important and that you have to understand the uh the underlying philosophy to exist as a functional member of society it's a way to do so and that's a way which the throne basically affects people to believe in the kind of exceptional qualities of the subbrain which are necessary for a state uh a state that has a sovereignty to exist so the throne in the crown or or any other kind of token there so it's um it's embodied it refers to a wearable or a bodily position it refers to of course in a cultured and a semiotic context it's enacted that relates to declarative speech acts i pronounce you married the speech act is the action and so when we're talking about writing and language driven systems which is a key part of again that that deep histories section declarative speech acts become very important when you're served the subpoena or when something is ruled then the hammer comes down on the table with like sort of like the finality of the act like an onomatopoeia relative to the declaration that was uttered in a certain ritualized context these are great examples about how we can use like the 4e framework um and also i was going to say it's extended because the crown and the throne they're literally off of the body and then what's the story that's like the total um like oh king got owned emperor's new clothes where the extended cognitive apparatus that signals the cultural position of the king is revealed to be not what it is

and so it's like to strip the person of their role it's just like stripping off their clothes and these are just all kinds of very accessible phenomena that we can ask how does this support a 4e or an active inference approach relative to some other approach like signal processing yeah i'm sure you can make the neural network that recognizes a crown versus other images but that is like a sub-component of visual perception within this broader picture and then also it helps us ask um what if and how we could design embodied and cultured enacted extended etc systems that reflect the statistical regularities of who we are now and how we think things are and how we expect and preferred things to change yeah i i think that's a very good point maybe this is where you know the survivor was talking about constraints there and also i suppose it comes back to the idea of when to stop i'm talking i talked about this a little bit with dean uh yes yesterday is um when do we know when to stop is one of the big challenges about you know ecological context and i keep thinking so you know you have the hammer comes down you know if you have a state you know the hammer comes down it you you you viscerally have an embodied understanding of what's being communicated by then you know that comes down something's been hammered out and in the houses of parliament in the uk they have this big mace they call it the mace it's an ornament an ornamental mace but of course amazed well i don't know if this is exactly true but i i think the mace normally as something you club someone at the head with right in a battle right it's there as maybe that that big stick con idea that's been brought in but why one thing that really came to mind when i was when this was coming up was what happened during nazi germany when the new technology of sound systems came out so the big thing that happened was these big rallies that completely overwhelm people is no one had had just like we didn't have with social media before no one had had an embodied experience to calibrate and in some ways you maybe never can because it becomes slightly beyond the concepts of embodiment what it meant to be in this big arena where you heard sound coming from everywhere and it then the idea is that that was part of the reason why it became disruptive maybe we weren't able to constrain and it just became over a cancerous almost overwhelm but there's something about new technologies and the ability to constrain and find a state that is sustainable um and when it becomes out of control and is there something about the embodiment that daniel was talking about there thank you steven serval and then anyone else so i want to emphasize that uh as in their formalism i'm a cleaner too the the constraint is not um does not entail the notion of an interdiction of a tap down order or of of a stop on a behavior a constraint is a through slow process that modulates that modulates a fast process for example if we agree on the sound of some word say a lamp

so we have a relatively slow process that is our couple behavior that will constrain our relative faster processes in the sense that when one of us use the word lamp the other will understand what it refers to so it will basically lower the dimensionality of the possible behavior i have because i will only pick behavior that are consistent with the fact that you said the word lamp as i understand it so the way that we culturally build our niches it's not something we understand as constraining it's something we understand as creative and yet it corresponds potentially to the um the concept of constraint as would be used by a physicist or stem biologist which is the sense i use in that paper thanks it reminds me of a paper could find it with like uh it was like developmental constraints or opportunities and it was just saying you can think about like okay i guess there's some morphological creature that just extends in a blob but any other one that has facilitated enabled affordances to grow in one can be framed as the constraints from the other dimension and so there are two sides of the coin to talk about what is the constraint that enables it's like chess it's the enabling constraints on chess moves that allow brilliant chess games and a culture to actually arise so survival with the direct response then stephen and blue yeah thank you uh i think blue uh other hand at first i mentioned uh but what i wanted to say is that um to have something that exists as a meaningful concept to have interesting uh you need essential breakings for example you need a particular gas to not be independent the default assumption of a physicist is nothing everything depends on nothing each particle evolve independently essentially this means that i need basically the mean and the variance of their individual trajectories to to have all of the insert information on my system uh this correspond uh to like pressure and temperature and depending on what the trajectory is well this there is no claimation actually uh whatever but if i uh somehow uh forbids um if i somehow build a collection between the behavior of those particles then there are new possible behavior but emerge you have new possible elements of this dead space that exist you have a creation as far as a physicist that is concerned with behavior within certain species um is as far as uh by phrase but the physicist sees question because there are things that were not part of the set base he drew at the beginning of this system that become an existence something that was not in that now is and what is interesting is this process us i don't know who uh as we discussed earlier uh because uh you have kinds of symmetries building and symmetry spracking that are constructive and yet they need to be constraining because if everything just arbitrarily adopts any kind of behavior in the space that is accessible that is possible in principle you have just a philanthropy you have what noise you have literally nothing there you have just particles that blah the facts uh of existence that we

have things to talk about it entails uh the concerning of the possible space to everything is acrylally to some of the things are more likely and it entails a meaningful map between the possible states at the macro level at the lowest possible level of description and the states as we are interested in in a specific uh at a specific moment so thank you i was uh writing on the way that constraints are not restrictive but creative thank you the absence of restraints statistically constraints would just refer to you know an unconnected bayesian graph just independently vibrating variables and so to have anything that's tracking or anticipating something else we're discussing the imposition of constraints using that term stephen and then andy you mentioned i said daniel you put that enabling systems uh slash constraints so i think there is that i i totally get and i think it's a very good point about you know you've got the statistics stochastic chaos and you've got the general dynamical processes and there's there's going to be some way of constraining that through different um there's also then uh you know if we take consciousness as being at the break the brain stem and not in the frontal cortex you know and it's then sending stuff out you know i can see that biological dynamic but you also have this enabling system process of scale i suppose that in a way i was just referring to and these things can start to be like the the ball rolling down the hill so to speak you know it starts to accumulate its own because it doesn't need to extract information to make inferences to make things happen it's just happening through thermodynamics and dynamics so um i wonder how we can think about the challenge of these macro systems then the enabling system overriding the emerging dynamical system andy is there a fundamental difference between thinking like a state and thinking like a tribe or is it just a level of sophistication um with what specialties are are enabled once you get a state level thanks for the question sir states states has administrative systems as a tribe does not like you are a lot of ways societies can be organized that are uh somewhat in a continuum and what i call a state is a specific uh control system that well is in a precipitatory state either it exists or it does not and in the case of uh tribes i don't think um most of what we call tribe of an integrated decision system more like shared ethnic cultural norms and if they have intuitive decision systems it's certainly nothing like uh administration it's so fascinating because the tribe can be implicitly existing without a state as cervical just noted and it makes me think about web 3 and taos where it's almost like the states are being defined first and then the people are rushing in to fill so instead of the historical emergence of specialization and elaboration formalization on top of an implicit background of the people and then in a complex interplay as we've been discussing this is almost like a computational state that people are migrating to with a preset governance system sometimes

one that's incredibly constraining and people may not even know how constraining until they're there so that's an inversion how we've seen the um members of the state and the state's evolution engage so more things change the more they stay the same but it's a different river for sure so dean blue stephen i think i found something that i can i can say i can agree with with serval and and maybe some of the people who really like genetics because i i think constraint really matters and i i think the idea that ratio is not phenotypical drift through uh neutral mutations even though mutations are a ratio of the non-mutations i also think that ratios and randomness accommodations are not the same thing so i think constraint allows us to be able to see the difference between those two things one is when a triangle is a triangle and the other is when a variation plays upon a variation and i think we can accept that's what i think the constraint piece actually helps us pull the two things together i thank you dean blue then stephen so just to kind of go back to what um andy was saying about the difference between a tribe and a state but also like when do we know when to stop um and processes that that constrain and when i think about this like bi-directional information flow like if the emergent energy is too strong like it then it's out of control right there's no constraint but but like when the constraint is too strong like i think about like you know even in mental action the idea that like you have your grip is too tight and so like what whatever you're trying to control like pops out of your grip right like so like you can squeeze a balloon and then boom um and so there has to be like some balance and i just wonder if that balance is like quantifiable like does downward information have to equal upward information flow like is there like a balance between those two things that allows like a self-emergent or not a self-organizing entity to maintain in the state of an attractor because if like the information flow is too much in one way the balance goes off and it'll steer away from that attractor space steven and then i'll add a comment on that blue yeah i think that there's the the ability to constrain is going to be at different levels in in you've got certain things which are just constitutional you hear this in the states there's constitutional rights right they're fixed they can't change it's been there for 200 years or whatever and then you've got more negotiation and in the uk you've got the government has to go and talk to the queen every couple of weeks or every week so in theory the queen could constrain now again they don't just have to send a letter they have to go and meet in person and talk there has to be so then there's this inter-subjective dynamical constraint and then of course then for scaling you've got the kind of we can't really do it as organisms so we put in an electoral system but the electoral system and the the democracy in north america comes a lot from the tribe that the the sort of scaled tribes of north america as

much as it does from france it's quite interesting and one thing they have in toronto we have the treaty of the dish of one spoon and the treated disher one spoon is basically they sat round in a circle and they have a spoon they have a dish with one spoon and you only take enough for yourself and you leave enough for nature and it's past rounds and that's the treaty of how to look after the great lakes and actually it was a lot of that kind slightly scaled version of tribal work which then informed and said okay europe you're killing each other all the time what about something a little bit like this so actually a democracy in the u.s was partly from the euro not really tribes anymore but that kind of scaled peace so i think um there's there's there's that need to sort of draw back into something which is more contextual more dynamical um as well as the kind of i don't know the more kind of law based approach thank you so people have been mentioning the getting a grip which was of course from the about how states get a grip on their subjects and that was drawn from the work of of yella and ritfield talking about optimal grip and skilled performance and all these other cool inactive and ecological psychological ideas we've talked to so it's like of all the trajectories that the golf club can take we're constraining that to some lower dimensional manifold that's getting a grip and i totally agree i think a good example of an empirical system and about what kind of reciprocal interactions exist and what attractor states does that beget uh is the federalist anti-federalist discussion at the end of the 1700s in the united states with people saying if you have the affordance for a standing army at the federal scale then we'll see this happen at the lower state level or if there's the ability to engage in foreign diplomacy or print debt at the federal level then we'll see this happen or not at the state level and so i think that's where we go from sort of the what has been and what is to the future that we don't know about the what-if we are in that scale-friendly level where we're talking about designing or modifying or perturbing or intervening in those causal threads that do link and co-constrain the smaller sub-units with the larger processes how do we talk about the space of design there and integrate our preferences and expectations together with just sometimes the um dazzling and tragic and complex history of just everything we see on this map so we'll have just um a few minutes left let's from anyone who would like to let's just hear like some last thoughts what's something cool from today or what's something that you're looking forward to discussing more next week and otherwise it's been an awesome dot one so stephen and then we'll go around to anyone else yeah following on what you were saying there daniel and i'll use this as a sum up it slightly opens up a new thing which maybe we'll get to the next session but you know how do we see what happens with the nature of tech companies and the huge affordances that that puts into

our landscape also the constraints if we take the mcluhan tetrad of how it then changes the way we behave overall they're introducing rather than um a constraining or a sort of a wise constraining way it's it's putting out there what's out there and it's it's affording huge amounts of new novel interactions be it microsoft be it google be it facebook and then they're getting hugely powerful just through permeation into society so uh you know something that's interesting is then where does you know i think that may reflect one of the challenges we have at the moment with corporations getting so big so thank you and this is a very very good talk and lots of things are sparking off it so i appreciate it thank you yeah like nation states used to be like states were the big players in town there was big ones and there were smaller ones but that was sort of like the ultimate level of analysis web 2 we saw websites stock tickers apple amazon etc become bigger than many states and intertwined and amongst them in a way that sometimes is even more advantageous though they don't have citizenships themselves they have user identifiers they have revenue flows they have governance models and then now into web3 we go and platforms and protocols maybe even decentralized ways of accumulating governance and economic that will also be at least initially larger than smaller nation states and web to tech companies so there's gonna be like legacy like web zero one two three four all of these systems are going to be on the same territory but they're going to have different maps so they're going to be coordinating very differently um would anyone else like to give a closing thought i think there's a lot of stuff in the dot too that we'll be able to like jump off into like survival next week i want to hear about where we go next like what are your next personal research goals and what are other opportunities for individuals who want to like learn more or research or apply in this area or under this framework blue and then anyone else so just a final thought i'd like to like discuss more or maybe get more into like what is the state anticipating i mean how do they anticipate uh what's coming next just predictive behavior yeah and it will be fun also to like just connect some of these to the core terms in the active in anthology just like what are the sense states of the states what are the action states of the states let's you know what is attention is that the most wanted list what are the ways that we can just take the terms that we already have and start to build up this table of mappings so that we can and maybe it's gonna fit on multiple levels so it's not gonna be just the one and only map of maps but it'll be a draft versionable participatory set of mappings and that'll be fun yeah what i really appreciated about the paper was the ability to sort of look at things through the scale in variance lens and now i'm really curious to see what would happen if we got off the map it started moving around and then actually incorporated scale variants because i think that's really the and then

what happens piece that this paper of paper forwards it gives us a good grounding and now we can step off of that and see to think in the hypothetical space i'm really curious to see where that goes excellent andy if you have any last comments well thanks for letting me play i really enjoyed the conversation um and a lot of this was new to me right i'm new to the even the idea of active inference but being able to discuss it in concepts of the state uh incredibly interesting so thanks for letting me participate excellent well it's always a two or more way street so we all had an awesome time thank you everybody for participating we hope to see you here and or participants through listening in another live stream in the lab working with us on projects so see you all later talk to you next time